

Exercise 3: Interpretation and Robustness Assessment

1. Which risk drivers dominate uncertainty in economic performance?

For the electrical subsystem, the main risk drivers are hardware failure, operating cost variability, and power/thermal mismatch. Hardware failure is the biggest risk because it has a high cost impact (\$52,500 per event) and a noticeable probability (13%). This leads to an expected loss of around \$6,800 per year, making it the largest contributor to uncertainty. Operating cost variability is also important since it happens frequently (75%). Changes in electricity prices and energy use can increase yearly costs and create ongoing uncertainty. Power and thermal mismatch has a moderate impact, with an expected loss of about \$4,500 per year ($25\% \times \$18,000$). This mainly comes from inefficient energy use or overheating issues. On the other hand, certification uncertainty has very little impact because its probability is low (2%), so it does not significantly affect overall economic performance.

For the software subsystem, cybersecurity update uncertainty and operating cost variability due to cloud and energy use are the two risk drivers that dominate uncertainty in economic performance, as they both carry high probability and high financial impact. Cybersecurity updates pose a 50% annual likelihood with a \$200,000 impact, yielding an expected annual loss of ~\$100,000, with a \$75,000/year residual exposure remaining even after mitigation. Operating cost variability occurs at 75% annually with a \$45,000 impact, producing ~\$33,750 in expected losses and acting as a persistent drag on the software subsystem's economic performance throughout the project horizon.

Hardware failure presents moderate exposure (~\$6,825/year) for the software subsystem, though its \$138,248 mitigation cost warrants scrutiny over shorter time horizons. Certification uncertainty contributes minimally, with a negligible ~\$800/year expected impact that is effectively eliminated through mitigation.

2. How do the sensitivity and Monte Carlo results compare?

For the software subsystem, the sensitivity analysis shows that number of cameras and storage capacity are the most influential variables. For the number of cameras, the PW ranges from \$3,393,630 (at +20%) to \$3,610,537 (at -20%), giving a total swing of ~\$217,000, or roughly ~\$108,000 from the base case in either direction. For storage, the PW ranges from \$3,405,234 (at +20%) to \$3,598,933 (at -20%), giving a total swing of ~\$194,000, or roughly ~\$97,000 from the base case in either direction. Network bandwidth has virtually no effect across the full $\pm 20\%$ range, indicating the design is robust to bandwidth fluctuations.

The Monte Carlo results complement this by capturing all risk drivers simultaneously. The stochastic mean PW of \$2,379,227 is significantly lower than the deterministic base case of \$3,502,084, confirming that the base case is optimistic once uncertainty is factored in. Despite this, the 100% probability of PW exceeding zero across all simulations confirms the software subsystem remains economically viable under uncertainty.

For the electrical subsystem, the sensitivity analysis shows that energy consumption and number of cameras have the biggest impact on economic performance. When energy consumption changes, operating costs change directly, which leads to a clear shift in PW. Similarly, increasing the number of cameras increases both capital costs and energy usage, which lowers overall economic performance. Compared to other variables, these two clearly drive most of the variation. The Monte Carlo results give a more complete picture because all risks are considered at the

same time. The stochastic mean PW is lower than the deterministic base case, which shows that the base case is slightly optimistic when uncertainty is included.

However, the probability of PW staying above zero is still close to 100%, which means the system is still economically feasible even under uncertainty. Also, the spread in the Monte Carlo results is wider than in the sensitivity analysis, showing that multiple risks happening together can increase overall uncertainty more than changing one variable at a time.

3. How robust are your conclusions to uncertainty and risks?

For the software subsystem, conclusions are moderately robust but limited in scope. Only four risks were included in the stochastic model, meaning unmodeled threats such as supplier dependency or newly discovered security flaws could materially shift economic outcomes. The two dominant drivers, cybersecurity update uncertainty and cloud/energy cost variability, were modelled with appropriate $\pm 50\%$ ranges, providing reasonable coverage of the most impactful risks. However, certification and hardware failure contributed minimal spread due to their narrow uncertainty ranges, meaning the true economic uncertainty is likely understated overall.

For the electrical subsystem, the conclusions are reasonably robust but still subject to some limitations. The stochastic model includes the major identifiable risks such as hardware failure, operating cost variability, and thermal mismatch, which capture the most significant sources of uncertainty. However, some potential risks are not explicitly modelled, including long-term component degradation, supply chain disruptions, or unexpected increases in energy prices beyond the assumed range. As a result, the current model may slightly underestimate total uncertainty. Despite these limitations, the relatively narrow confidence interval and consistently positive PW across all simulations indicate that the electrical subsystem design is stable and resilient under the considered risk conditions.

4. What are the effects of the mitigation strategies? Which ones should you implement?

For the software subsystem, the most cost-effective mitigation strategy to implement is reserved pricing, FinOps reviews, and real-time monitoring, as it delivers a 35% reduction in unexpected operating costs at a relatively low implementation cost of ~\$6,000/year using native cloud tools. This makes it the highest value-to-cost strategy among all four risks.

The cybersecurity update mitigation, hiring a dedicated expert, should be considered despite its higher cost of \$104,224/year, as it targets the single largest risk driver, reducing its probability by 25%; unmitigated exposure reaches ~\$100,000/year in expected losses. At this stage, it should not be implemented as the costs slightly outweigh the benefits. However, additional research on this mitigation should be done to determine the exact costs and benefits and decide whether it is worth implementing.

For hardware, N+1 redundancy should be applied selectively to critical components only, such as servers and core networking infrastructure, rather than all hardware. This limits capital expenditure while still protecting against the most consequential failure points, reducing downtime and future labour costs. Certification mitigation through early planning and schedule buffering should also be implemented, as its cost is primarily staff time and delivers reliable protection against a risk that, if realized, could delay the entire project go-live.

For the electrical subsystem, uncertainty shows that the design is most vulnerable to energy-related costs and hardware reliability. These factors directly affect both operating and maintenance costs, making them the main sources of risk. The system is least vulnerable to certification-related risks, since they have very low probability and limited financial impact. Overall, the design performs well under normal conditions, but could face higher costs if energy prices increase significantly or if hardware failures happen more frequently than expected.

5. What does uncertainty reveal about where the Primary Design is most and least vulnerable?

Uncertainty reveals that the software subsystem's primary design is most vulnerable to operating cost variability and cybersecurity update uncertainty, as both combine high likelihood with high financial impact. This is evident in the stochastic model, where refreshing the simulation produces PW and AW swings in the hundreds of thousands of dollars, reflecting how sensitive the design's economic performance is to these two drivers.

The design is least vulnerable to certification timeline uncertainty and hardware failure, as both contribute minimal spread to economic outcomes due to their low probability or narrow uncertainty ranges.

Overall, the primary design performs reasonably well under average conditions, but is exposed to significant downside risk in years where dominant risks materialize more frequently than expected. Conversely, favorable years where these risks occur rarely could meaningfully improve economic outcomes.

Exercise 4: Interpretation and Robustness Assessment

For the software subsystem, during week 7 we selected the design that focused on redundant security systems and cybersecurity. That is the design shown in the software subsystem excel file. We selected this design due to its high magnitude benefits resulting in a higher PW and AW (a more complete comparison with values can be seen in the Week 7 document). Additionally, the chosen software subsystem design has a lower IRR, however it works on a much larger scale than the alternative design (which was focused on easy control of the software). It was decided that the decreased IRR in return for higher PW and AW was worth it, hence our decision. Now, incorporating risk framing and utilizing the sensitivity and Monte Carlo results, the PW and AW are still much higher for the redundant and cybersecure design so our decision has not changed. However, as explained in Exercise 3, part 4, there are some mitigation strategies we will be adding to the design to lessen the likelihood / impact of some of the more pertinent risks.

For the electrical subsystem, the design selected in week 7 focuses on ensuring stable power delivery and efficient energy usage across the system. This design was chosen because it provides a strong balance between capital cost and long-term operating cost, resulting in a positive PW and AW compared to alternative configurations. From the deterministic analysis, the selected design shows consistent economic performance, with operating costs remaining manageable due to controlled energy consumption and system efficiency. Although alternative designs may reduce initial cost, they result in higher long-term energy usage, which lowers overall economic performance. From the risk framing in Week 9, the main concerns for the electrical subsystem were hardware failure and operating cost variability.

The chosen design performs well under these risks, as it limits exposure through improved system efficiency and reasonable hardware reliability. Based on the sensitivity analysis, energy consumption and number of cameras were identified as the most important variables. The selected design maintains these variables at a level that avoids excessive cost increases, making the system less sensitive to fluctuations. The Monte Carlo results further support this decision, as the stochastic PW remains positive across all simulations, even when uncertainty is included. Although the average PW is slightly lower than the deterministic case, the system still performs consistently well and does not show significant downside risk. Therefore, the recommended electrical subsystem design is to maintain the current configuration, while implementing targeted mitigation strategies such as power optimization and selective hardware redundancy to further reduce risk and improve reliability.